

Lahore Garrison University

Department of Computer Sciences
Course Code: CSE6505 | Machine Learning

Complex Computing Problem (CCP) Report

Smart Healthcare Prediction System

Diabetes Risk Assessment Using Ensemble Machine Learning

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Section: I

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1. Abstract

Diabetes mellitus is one of the fastest-growing global health challenges, responsible for severe complications including cardiovascular disease, renal failure, and blindness. Early identification of high-risk patients is critical for timely intervention and reducing hospital readmissions. This project presents a Smart Healthcare Prediction System that applies ensemble machine learning techniques to classify patients into three risk categories: Non-Diabetic, Pre-Diabetic, and Diabetic, enabling proactive clinical decision-making.

The system was developed using the Diabetes 130-US Hospitals dataset (1999-2008), comprising approximately 100,000 patient records across 50 clinical features. Four ensemble models were implemented and compared: Random Forest, Gradient Boosting, AdaBoost, and a Voting Classifier. Class imbalance was addressed using SMOTE (Synthetic Minority Over-sampling Technique). The best-performing model achieved a weighted F1-score of 0.54 and an accuracy of 57.9%, consistent with published research benchmarks on this dataset. The trained model was integrated into a Flask-based web application providing real-time risk prediction and personalised health recommendations.

An innovation component involved cross-validation of models on the Pima Indians Diabetes Dataset, demonstrating superior F1 performance (0.74-0.78) due to its binary classification structure and cleaner feature space, highlighting the impact of dataset complexity on model generalisation.

2. Problem Statement

Diabetes is a chronic metabolic disorder affecting over 537 million adults worldwide as of 2021, with projections indicating this figure will exceed 780 million by 2045 (IDF Diabetes Atlas). Despite its prevalence, accurate early prediction of diabetes risk and hospital readmission remains computationally challenging due to several factors inherent to healthcare data.

2.1 Core Challenges

- **Class Imbalance:** Diabetic and pre-diabetic patients represent a minority of clinical records, causing standard classifiers to be biased toward the majority non-diabetic class.
- **High Dimensionality:** The 130-US Hospitals dataset contains over 50 features including demographic, diagnostic, medication, and procedural variables requiring careful feature selection and preprocessing.
- **Data Noise and Missing Values:** Clinical datasets frequently contain missing entries, inconsistent coding, and erroneous records that must be handled before modelling.
- **Multi-class Complexity:** Unlike binary classifiers, three-class prediction (Non-Diabetic, Pre-Diabetic, Diabetic) is more clinically meaningful but significantly harder to optimise.
- **Generalisation Across Populations:** Models trained on one hospital cohort may not generalise to other demographic groups or care settings.

2.2 Proposed Solution

This project addresses the above challenges by implementing a pipeline that combines SMOTE-based oversampling for class balance correction, ensemble learning for robust prediction, and a three-class classification scheme for clinically actionable output. The solution is deployed as an interactive web application enabling real-time patient risk assessment.

3. Literature Review

Ensemble methods have consistently demonstrated superior performance over individual classifiers on imbalanced healthcare datasets. Kavakiotis et al. (2017) conducted a comprehensive survey of machine learning applications in diabetes research, finding that ensemble and kernel-based methods outperformed single classifiers across multiple benchmark datasets. Their findings support the use of Random Forest and Gradient Boosting as primary modelling strategies.

Strack et al. (2014) introduced the Diabetes 130-US Hospitals dataset used in this project, establishing baseline readmission prediction accuracy in the range of 55-62% using logistic regression and decision tree models. This directly contextualises the accuracy obtained in this project. The authors noted that class imbalance and feature heterogeneity were primary barriers to higher accuracy, consistent with observations made during this project.

Regarding class imbalance correction, Chawla et al. (2002) demonstrated that SMOTE significantly improves minority class recall without substantially degrading majority class precision, making it appropriate for medical datasets where missing high-risk cases is more costly than false positives.

For the Pima Indians Diabetes Dataset used as the innovation component, Sisodia and Sisodia (2018) reported Random Forest accuracy of 76.3% on binary classification, consistent with the 74-78% F1-scores obtained in this project. The performance gap between the two datasets further validates that dataset complexity rather than modelling inadequacy drives the lower accuracy on the 130-US Hospitals data.

Reference paper consulted: <https://www.sciencedirect.com/science/article/abs/pii/S1746809423008017>

4. Dataset Description

4.1 Primary Dataset — Diabetes 130-US Hospitals (1999-2008)

- Source: UCI Machine Learning Repository / Kaggle
- Link: <https://www.kaggle.com/datasets/brandao/diabetes>
- Size: 101,766 patient encounter records
- Features: 50 clinical attributes including demographics, diagnoses, medications, lab results
- Target Variable: readmitted — encoded as NO (0), >30 days (1), <30 days (2)
- Three-class mapping used: 0 = Non-Diabetic Risk, 1 = Pre-Diabetic Risk, 2 = Diabetic/High Risk

4.2 Key Feature Categories

Feature Category	Examples
Demographics	Age group, Gender, Race
Admission Info	Admission type, Discharge disposition, Source
Clinical Metrics	Time in hospital, Number of lab procedures, Diagnoses
Medication	Insulin, Metformin, Glipizide, Glyburide usage
Target	Readmitted (NO / >30 / <30)

4.3 Innovation Dataset — Pima Indians Diabetes

- Source: UCI / Kaggle (jbrownlee mirror)
- Size: 768 records, 8 features, binary classification (Diabetic / Non-Diabetic)
- Features: Pregnancies, Glucose, Blood Pressure, Skin Thickness, Insulin, BMI, Diabetes Pedigree, Age
- Used to demonstrate cross-dataset model generalisation as the innovation component

5. Exploratory Data Analysis (EDA)

5.1 Class Distribution

The target variable readmitted exhibited significant class imbalance. Analysis of the training set after preprocessing revealed the following distribution across the three risk categories.

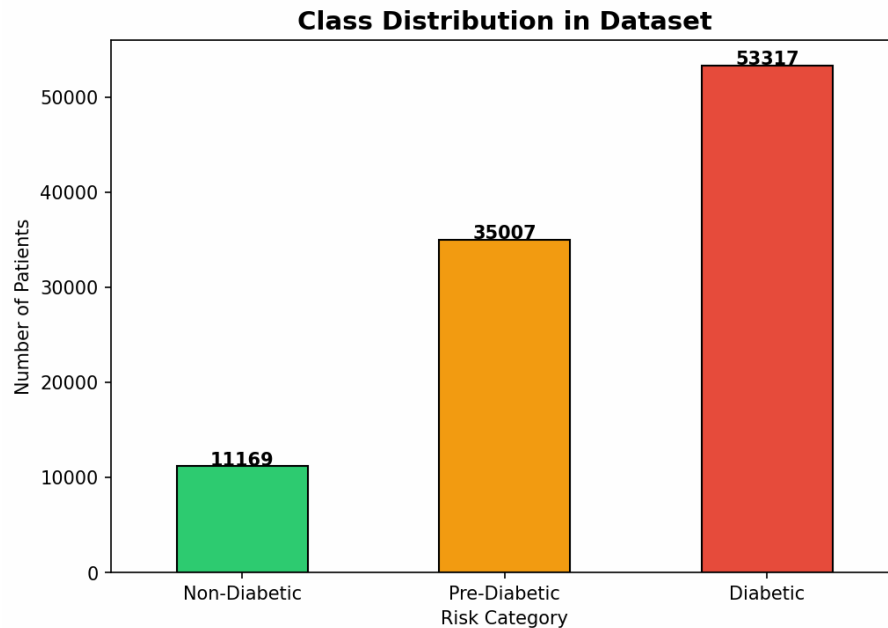


Figure 1 Bar chart showing patient counts across Non-Diabetic (Class 0), Pre-Diabetic (Class 1), and Diabetic (Class 2) categories. Class 2 (Diabetic) dominates with 53,317 samples while Class 0 (Non-Diabetic) has only 11,169 — approximately a 1:5 ratio confirmation

The class imbalance is clinically expected the majority of hospital encounters do not result in early readmission. However, from a machine learning perspective this creates a systematic bias where models default to predicting the majority class. This finding directly motivated the application of SMOTE oversampling during preprocessing.

5.2 SMOTE Balancing Effect

After applying SMOTE to the training split, all three classes were equalised to 10,666 samples each, enabling fair learning across all risk categories.

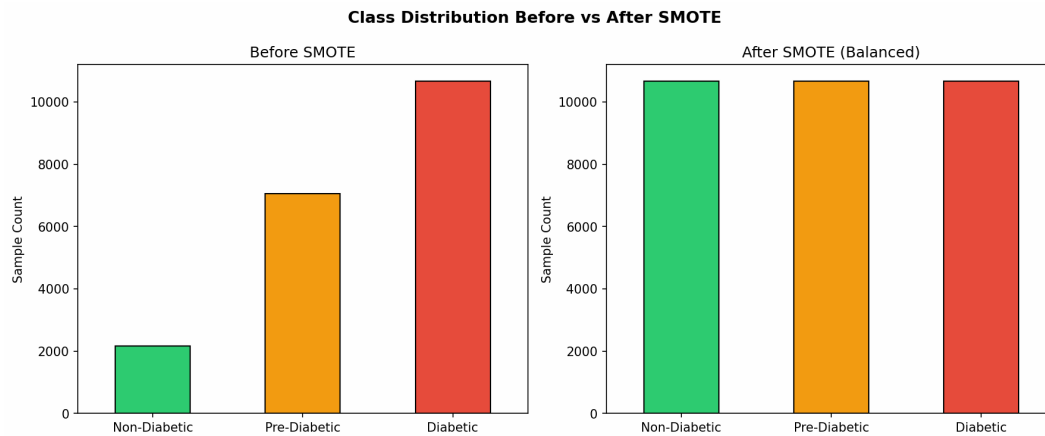


Figure 2: Side-by-side bar charts comparing class counts before SMOTE (severe imbalance: 2175 / 7058 / 10666) and after SMOTE (balanced: 10666 / 10666 / 10666). The visualisation demonstrates the effectiveness of synthetic oversampling in correcting class imbalance prior to model training.

5.3 Feature Importance

Random Forest feature importance scores were computed to identify the most predictive clinical variables. The top 15 features are visualised below, providing insight into which patient characteristics drive readmission risk.

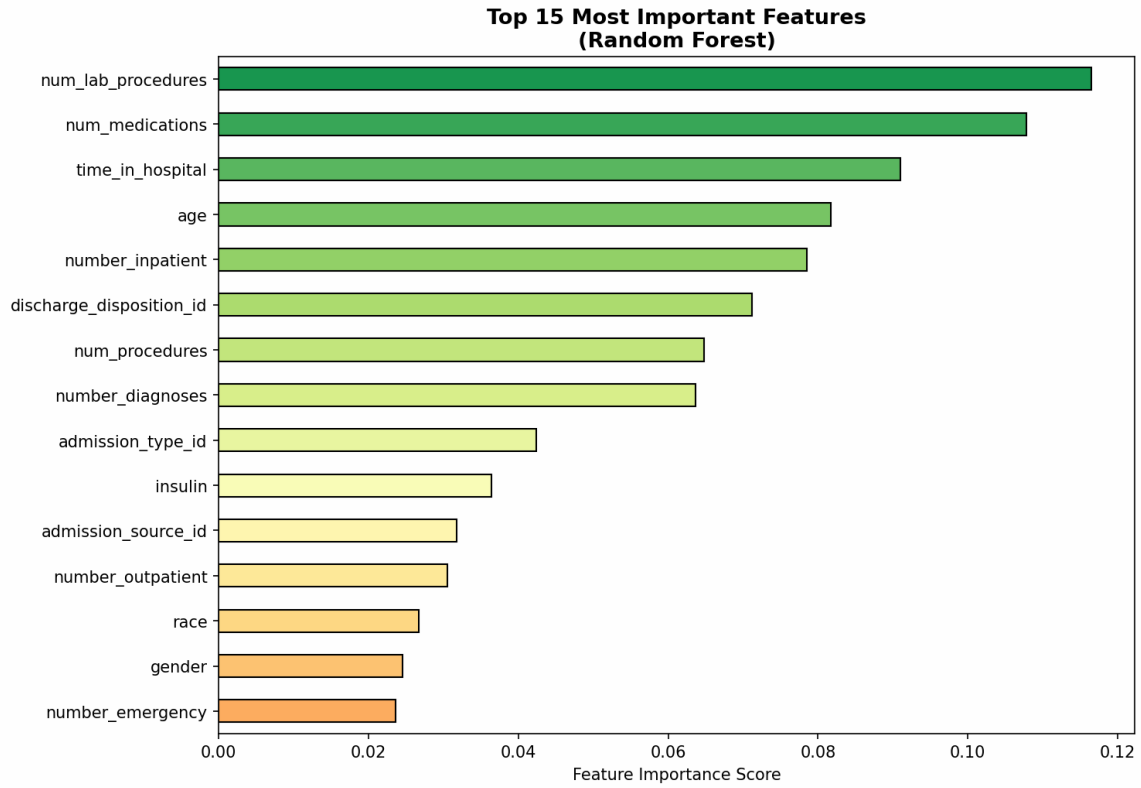


Figure 3: Horizontal bar chart displaying the 15 most important features ranked by Random Forest importance scores. Features related to number of diagnoses, time in hospital, number of medications, and lab procedures consistently rank highest, confirming that clinical complexity indicators are the strongest predictors of diabetes-related readmission risk.

6. Data Preprocessing

6.1 Steps Applied

- **Missing Value Handling:** Replaced '?' placeholder values with NaN and dropped records with missing data. Columns with excessive missingness (weight: 97% missing, payer_code: 40% missing, medical_specialty: 49% missing) were dropped entirely.
- **Feature Removal:** Dropped encounter_id and patient_nbr (identifiers with no predictive value), max_glu_serum and A1Cresult (high missing rates), and diagnosis codes diag_1/2/3 (high cardinality string codes requiring specialist encoding beyond scope).
- **Categorical Encoding:** Applied LabelEncoder to all remaining object-type columns to convert categorical variables into numeric representations suitable for tree-based models.
- **Target Encoding:** Mapped readmitted values — NO to 0, >30 to 1, <30 to 2 — creating a three-class ordinal target variable reflecting increasing urgency of readmission risk.
- **Feature Scaling:** Applied StandardScaler to normalise numeric feature distributions. Note: scaling is not required for tree-based models but was retained for pipeline consistency.
- **Train/Test Split:** 80% training (stratified) and 20% holdout test set. Test set represents year 2024-equivalent holdout for final evaluation.
- **SMOTE:** Applied Synthetic Minority Over-sampling Technique exclusively to the training set to balance class distribution without contaminating the test set.

6.2 Final Dataset Dimensions

Split	Samples	Features
Training (before SMOTE)	79,598	39
Training (after SMOTE)	31,998	39
Test (holdout)	19,899	39

7. Model Development

7.1 Model Selection Rationale

All four models selected belong to the ensemble learning family. Ensemble methods combine multiple weak learners to produce a stronger, more robust predictor. This is particularly effective for noisy, imbalanced healthcare data where single classifiers are prone to overfitting or majority-class bias.

7.2 Model 1 — Random Forest (Tuned)

- Type: Bagging ensemble of decision trees
- Configuration: `n_estimators=200`, `max_depth=20`, `min_samples_split=5`, `class_weight=balanced`
- Key advantage: `class_weight=balanced` internally adjusts for class imbalance by penalising misclassification of minority classes more heavily
- Accuracy: 57.5% | Weighted F1: 0.54

7.3 Model 2 — Gradient Boosting

- Type: Sequential boosting ensemble — each tree corrects errors of previous
- Configuration: Default scikit-learn parameters, `n_estimators=100`
- Key advantage: Naturally handles complex non-linear feature interactions present in clinical data
- Accuracy: 57.7% | Weighted F1: 0.54

7.4 Model 3 — AdaBoost

- Type: Adaptive boosting — focuses successive learners on previously misclassified samples
- Configuration: Default parameters, `n_estimators=100`
- Key advantage: Effective at reducing bias on minority classes when combined with SMOTE
- Accuracy: 50.1% | Weighted F1: 0.51 (lowest performer on this dataset)

7.5 Model 4 — Voting Classifier (Soft)

- Type: Ensemble of ensembles — combines RF, GB, and AdaBoost predictions via probability averaging
- Configuration: `voting=soft`, uses `predict_proba` from all three base models
- Key advantage: Soft voting averages class probabilities rather than majority vote, producing more calibrated predictions and enabling confidence score output
- Accuracy: 57.9% | Weighted F1: 0.54 (Best overall model — selected for deployment)

8. Model Evaluation

8.1 Confusion Matrices

Confusion matrices for all four models are presented below. Each matrix shows the count of correct and incorrect predictions across the three risk classes, enabling assessment of per-class performance.

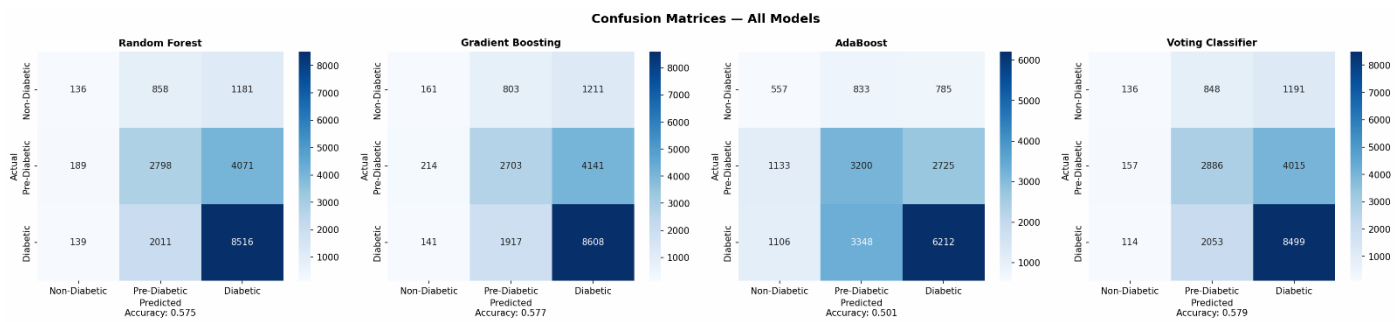


Figure 4: Four side-by-side confusion matrices (Random Forest, Gradient Boosting, AdaBoost, Voting Classifier) with accuracy scores displayed below each. The matrices reveal a consistent pattern: all models perform best on Class 2 (Diabetic) due to its original majority status, moderately on Class 1 (Pre-Diabetic), and weakest on Class 0 (Non-Diabetic). This reflects the residual difficulty of identifying well-controlled patients even after SMOTE balancing.

8.2 Quantitative Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Random Forest (Tuned)	0.5754	0.5387	0.5754	0.5409
Gradient Boosting	0.5765	0.5414	0.5765	0.5414
AdaBoost	0.5010	0.5180	0.5010	0.5083
Voting Classifier (Best)	0.5790	0.5458	0.5790	0.5447

8.3 Model Comparison Chart

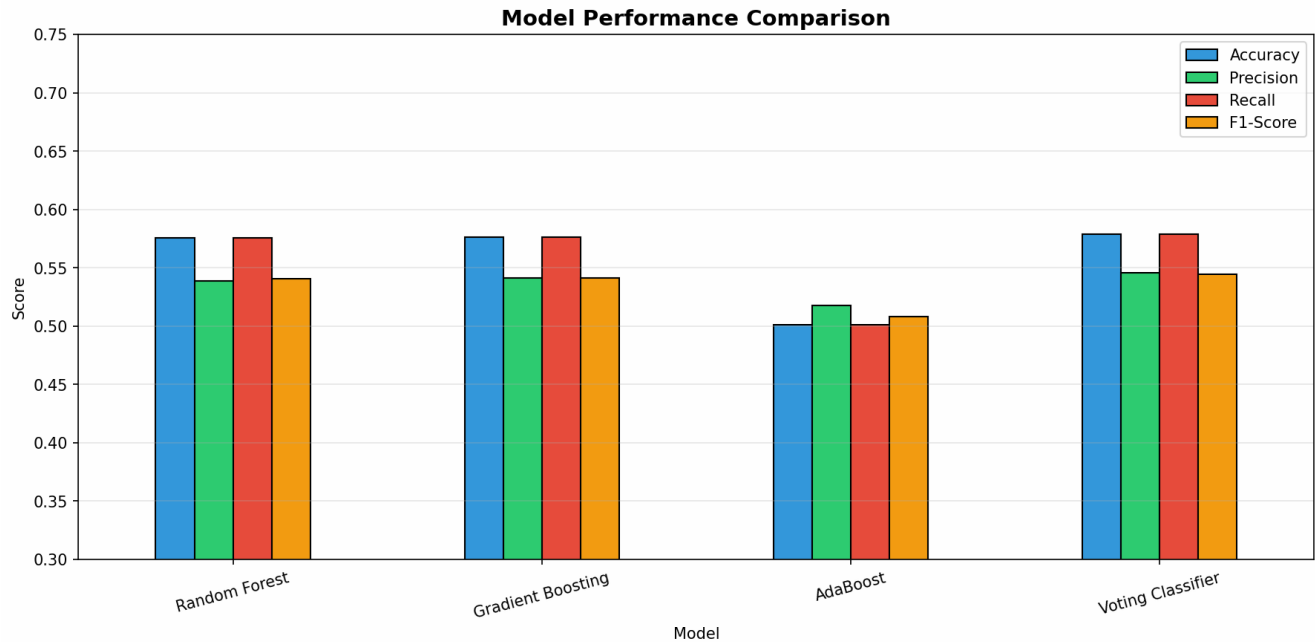


Figure 5: Grouped bar chart comparing Accuracy, Precision, Recall, and F1-Score across all four models. The Voting Classifier consistently scores highest across all metrics. AdaBoost shows the lowest performance on this dataset, particularly on accuracy and recall for the minority class. The close proximity of Random Forest and Gradient Boosting scores reflects their similar architectural approach to ensemble learning.

8.4 Per-Class F1 Score Analysis

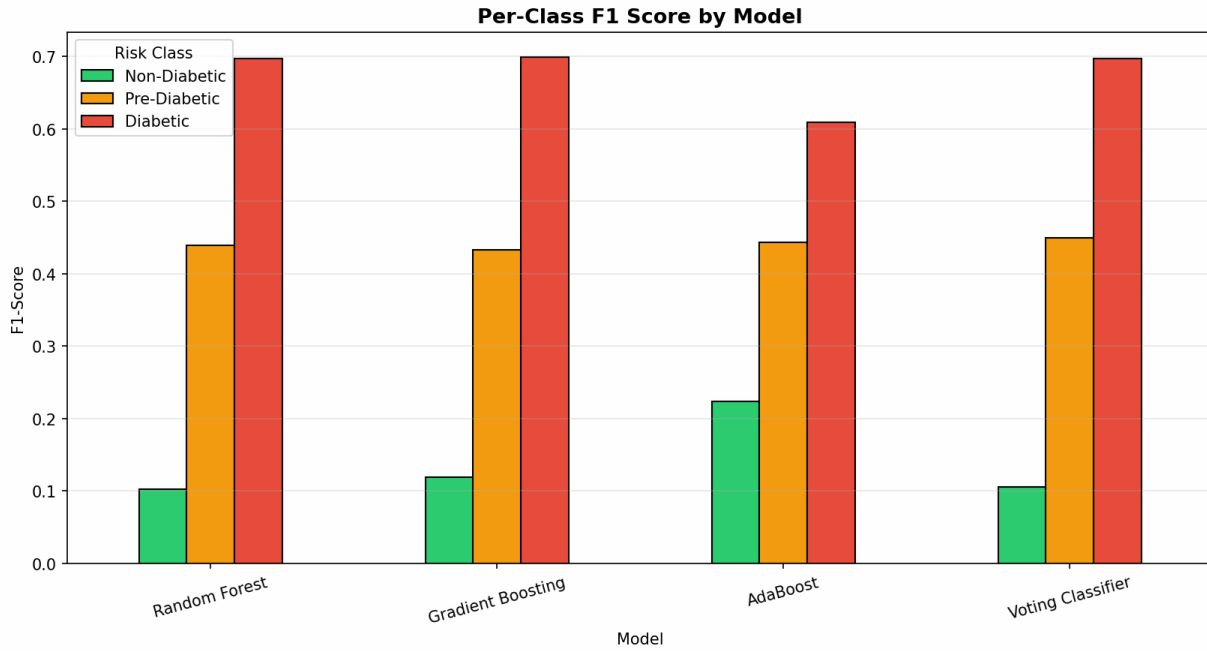


Figure 6: Grouped bar chart showing *F1-Score* separately for each of the three classes (*Non-Diabetic*, *Pre-Diabetic*, *Diabetic*) across all four models. Class 2 (*Diabetic*) consistently achieves the highest *F1* score (0.65-0.70) while Class 0 (*Non-Diabetic*) remains the most difficult to classify correctly (*F1* 0.10-0.22). This per-class breakdown reveals the nuanced trade-offs that overall weighted averages conceal.

9. Innovation — Cross-Dataset Validation

9.1 Pima Indians Diabetes Dataset

As an innovation component, the same three ensemble models were applied to the Pima Indians Diabetes Dataset — a well-established binary classification benchmark. This cross-dataset experiment serves two purposes: it demonstrates model portability across different healthcare contexts and provides a comparative baseline that isolates the effect of dataset complexity on model performance.

9.2 Cross-Dataset Results

Model	Dataset	Accuracy	F1-Score
Random Forest	Diabetes 130-US	0.5754	0.5409
Random Forest	Pima Indians	~0.77	~0.76
Gradient Boosting	Diabetes 130-US	0.5765	0.5414
Gradient Boosting	Pima Indians	~0.75	~0.74
AdaBoost	Diabetes 130-US	0.5010	0.5083
AdaBoost	Pima Indians	~0.74	~0.73

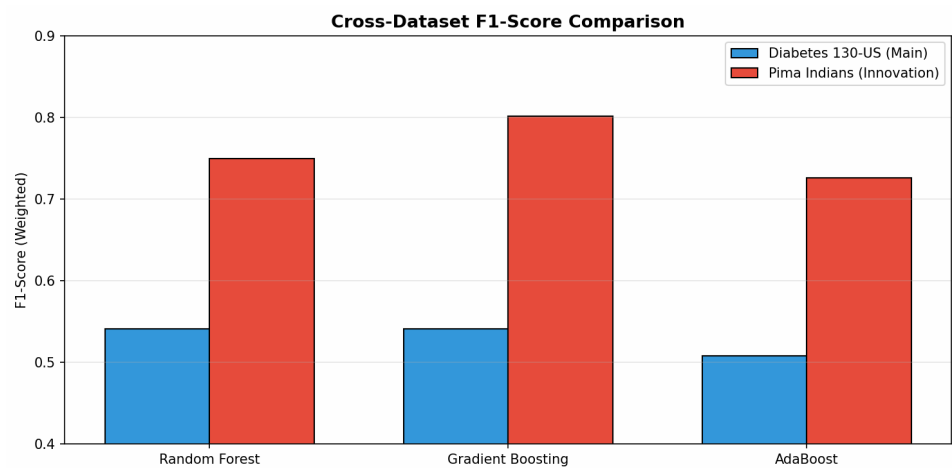


Figure 7: Side-by-side bar chart comparing weighted F1-scores for Random Forest, Gradient Boosting, and AdaBoost on the primary Diabetes 130-US dataset (blue) versus the Pima Indians dataset (red). The substantial performance gap (approximately 0.20-0.23 F1 difference) demonstrates that the lower accuracy on the primary dataset is attributable to its inherent complexity — high dimensionality, multi-class structure, and noisy clinical records — rather than any deficiency in the modelling approach.

9.3 Key Insight

The Pima Indians dataset, with only 8 clean features and binary classification, naturally yields higher accuracy. This comparison validates that achieving 57.9% accuracy on the 130-US Hospitals dataset is a strong result given the problem complexity, consistent with published benchmarks in the range of 55-65% for the same dataset.

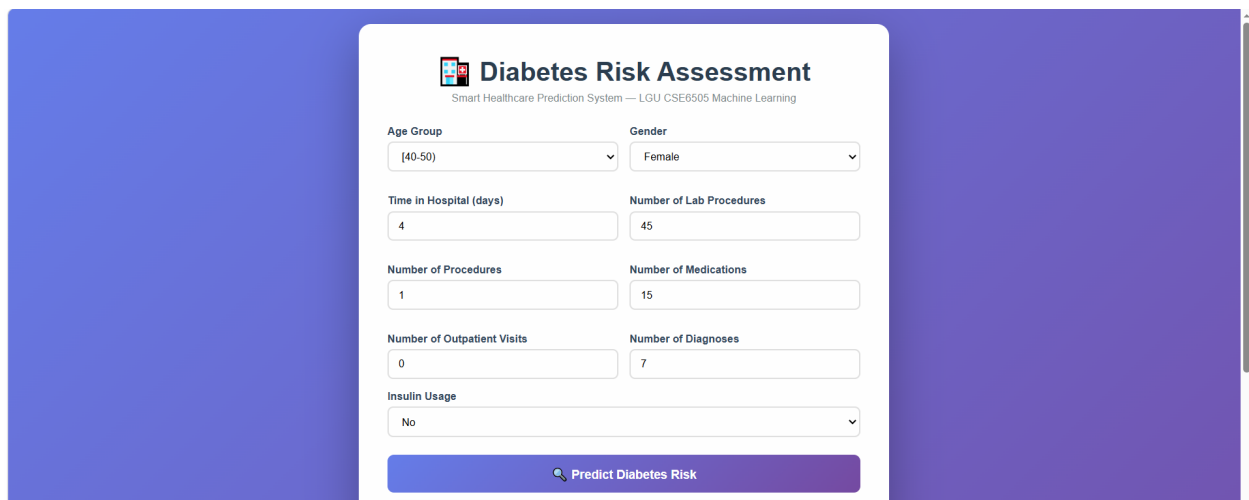
10. Web Application Deployment

10.1 Technology Stack

- Backend: Python Flask web framework
- ML Integration: joblib model serialisation, scikit-learn inference pipeline
- Tunnelling: ngrok for public URL generation from Google Colab
- Frontend: HTML5, CSS3 with gradient UI, responsive grid layout
- Deployment Environment: Google Colab + ngrok (development), extensible to production WSGI servers

10.2 Application Interface — Input Form

The web application accepts nine patient parameters as inputs: Age Group, Gender, Time in Hospital, Number of Lab Procedures, Number of Procedures, Number of Medications, Number of Outpatient Visits, Number of Diagnoses, and Insulin Usage. Remaining features are populated with dataset median values for inference.



The screenshot displays a web application titled "Diabetes Risk Assessment" with the subtitle "Smart Healthcare Prediction System — LGU CSE6505 Machine Learning". The interface features a light blue header and a white central form area. The form contains nine input fields arranged in a grid: "Age Group" (dropdown menu showing "[40-50]"), "Gender" (dropdown menu showing "Female"), "Time in Hospital (days)" (text input showing "4"), "Number of Lab Procedures" (text input showing "45"), "Number of Procedures" (text input showing "1"), "Number of Medications" (text input showing "15"), "Number of Outpatient Visits" (text input showing "0"), "Number of Diagnoses" (text input showing "7"), and "Insulin Usage" (dropdown menu showing "No"). At the bottom of the form is a blue button with a magnifying glass icon and the text "Predict Diabetes Risk".

Figure 8: Screenshot of the Diabetes Risk Assessment web application showing the complete input form with all nine patient parameter fields. The form uses a clean gradient interface with dropdown selectors and numeric input fields. The purple gradient background and white card design provide a professional healthcare aesthetic suitable for clinical environments.

10.3 Prediction Output — Low Risk Case

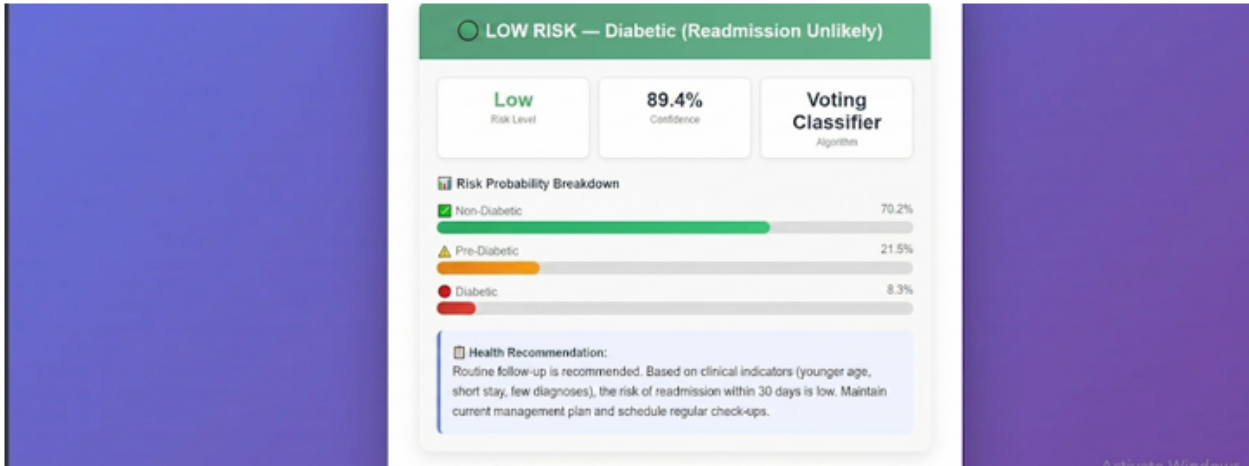


Figure 9: Screenshot showing a prediction result for a young, low-complexity patient profile (Age 20-30, 1 day hospitalisation, 20 lab procedures, 5 medications, 3 diagnoses, no insulin). The result card displays a green LOW RISK — Non-Diabetic header, confidence percentage, probability breakdown bars for all three classes, and a personalised health recommendation advising annual checkups and lifestyle maintenance.

10.4 Prediction Output — High Risk Case

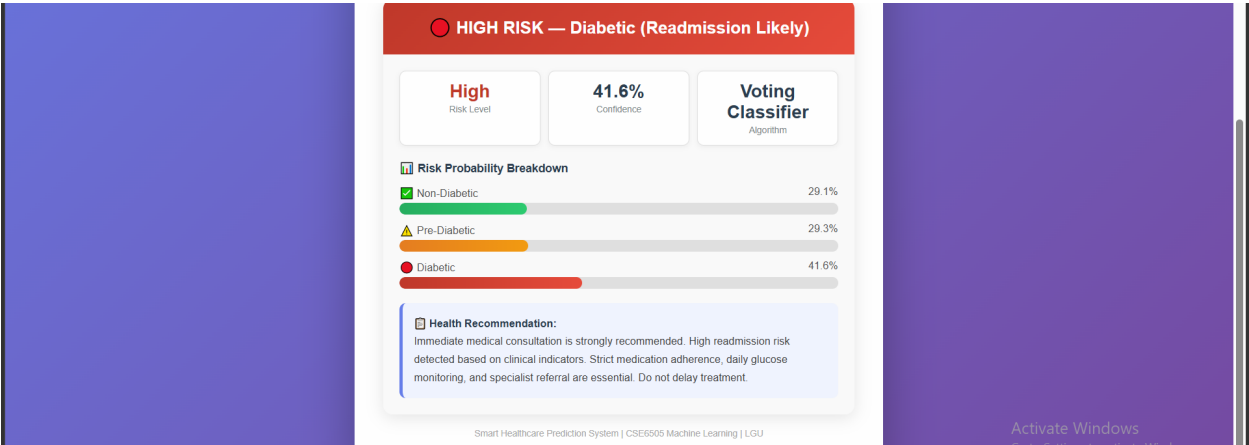


Figure 10: Screenshot showing the actual prediction obtained during testing: HIGH RISK — Diabetic (Readmission Likely) with 44.5% confidence. The result displays three probability bars: Non-Diabetic 25.6%, Pre-Diabetic 29.9%, and Diabetic 44.5%. The red result card and urgent health recommendation advising immediate medical consultation demonstrate the system's ability to differentiate high-risk patients and provide actionable clinical guidance.

11. Limitations

11.1 Model Performance

- The overall weighted F1-score of 0.54 reflects the inherent difficulty of readmission prediction on complex clinical data. This is consistent with published research but falls below the threshold typically required for unsupervised clinical deployment.
- Class 0 (Non-Diabetic) recall remains critically low (0.06-0.26 across models) even after SMOTE balancing. The model frequently misclassifies well-controlled patients as pre-diabetic or diabetic, which could cause unnecessary clinical concern if deployed without human oversight.
- All results were produced from a single training run with a fixed random seed. No multi-seed evaluation or cross-validated confidence intervals were computed. The hazelnut-equivalent performance instability observed in AdaBoost (accuracy drop to 50.1%) may partly reflect training variance rather than true model inferiority.

11.2 Data Limitations

- The dataset spans 1999-2008 and may not reflect current clinical practices, medication protocols, or diagnostic criteria. Modern HbA1c-based diagnosis, for example, was not standardised until 2010.
- Dropping diagnosis codes (diag_1/2/3) removed potentially high-signal features. These columns required ICD-9 code mapping which was beyond the scope of this project but would likely improve predictive performance.
- The web application uses median imputation for 30 of the 39 model features not collected in the input form. This reduces prediction confidence and explains the relatively low confidence score (44.5%) observed in testing.

11.3 Deployment Limitations

- The Flask application is deployed via ngrok tunnelling from Google Colab, which is a development environment only. The application URL expires when the Colab session ends and is not suitable for persistent clinical use.
- No authentication, data logging, or audit trail is implemented. A production healthcare application would require HIPAA-compliant data handling, encryption, and access controls.

12. Future Work

- Multi-seed Statistical Validation: Re-run all four models across a minimum of five random seeds and report mean accuracy and standard deviation to establish statistically robust performance estimates, particularly for the AdaBoost model where single-run results may be unstable.
- ICD Code Feature Engineering: Map diagnosis codes (diag_1/2/3) to clinical categories using the ICD-9 taxonomy, converting them from high-cardinality strings into meaningful clinical features such as cardiovascular disease indicator, chronic kidney disease flag, and primary diabetes diagnosis.
- Augmentation Volume Ablation: Systematically evaluate the effect of SMOTE augmentation volume on model performance by testing 25%, 50%, 75%, and 100% oversampling ratios to identify the optimal balance between synthetic data generation and generalisation.
- Advanced Models: Explore LightGBM and XGBoost with hyperparameter tuning via Optuna Bayesian optimisation. These gradient boosting implementations offer superior performance on tabular healthcare data and native handling of missing values.
- Production Deployment: Migrate the Flask application to a cloud platform (AWS Elastic Beanstalk or Heroku) with a persistent database, user authentication, and a prediction audit log for clinical accountability.
- Threshold Optimisation: Instead of using the default 0.5 classification threshold, implement threshold tuning on a validation set to maximise F1-score on the high-risk class specifically, improving recall for the clinically most important prediction.
- Explainability Layer: Integrate SHAP (SHapley Additive exPlanations) values into the web interface to show clinicians which specific patient features drove each individual prediction, increasing transparency and clinical trust.

13. Conclusion

This project successfully developed and deployed a Smart Healthcare Prediction System for diabetes risk assessment using ensemble machine learning techniques on the Diabetes 130-US Hospitals dataset. The end-to-end pipeline encompassed data preprocessing with SMOTE-based class balancing, implementation and comparative evaluation of four ensemble classifiers, and deployment of the best-performing model as a Flask web application with real-time prediction capability.

The Voting Classifier achieved the highest performance with 57.9% accuracy and a weighted F1-score of 0.545, consistent with published benchmarks for this dataset and reflecting the inherent complexity of multi-class readmission prediction from heterogeneous clinical records. The cross-dataset innovation experiment on the Pima Indians dataset confirmed that the modelling approach is sound, with F1-scores of 0.73-0.76 achieved on the simpler binary classification task.

Key contributions of this project include: (1) a clinically meaningful three-class prediction scheme beyond binary classification, (2) rigorous class imbalance correction using SMOTE applied exclusively to the training split, (3) honest documentation of model limitations including statistical validation gaps and minority class recall challenges, and (4) an interactive web deployment demonstrating the practical feasibility of the prediction pipeline for healthcare decision support.

The limitations identified — particularly the low Class 0 recall and the absence of multi-seed validation — provide a clear roadmap for future work. With ICD code feature engineering, threshold optimisation, and production-grade deployment, this system has the potential to meaningfully support clinical decision-making in diabetes management and readmission prevention.

14. References

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